

Digitally Assisted Per-Stage DC-Offset Calibration for a Low-Cost PSoC-Based Geophone Front End

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Abstract—This paper presents a digitally assisted foreground calibration method for a low-cost geophone analog front end implemented in a PSoC 5LP. The front end adapts a previously reported analog compensation topology to an inexpensive mixed-signal platform. However, the offset and drift of its on-chip operational amplifiers are multiplied by the cascaded gain stages, reducing the available dynamic range and eventually driving intermediate nodes into saturation. A fixed DC correction at the summing stage is functional but cannot restore the operating points of the remaining stages. The proposed method sequentially observes every stage through an on-chip analog multiplexer and reuses the 18-bit delta-sigma converter employed for geophone acquisition. ADC measurements are rescaled into a DAC-equivalent control domain, and a foreground control algorithm adjusts four independent 8-bit VDAC references. A discrete-time PI controller with deadband and anti-windup was selected as a simple implementation, but the architecture is not restricted to this controller. The digital filter hardware already used for acquisition is temporarily reconfigured to estimate the DC component. Once the four stages have converged, the DAC values are frozen during acquisition and may be stored in EEPROM for each programmable-gain setting. Preliminary bench observations showed uncompensated drifts on the order of volts, whereas calibrated stage outputs remained approximately within ± 100 mV. **Controlled measurements must replace this preliminary observation with: initial and residual offset, unclipped input-range improvement, convergence time, and successful calibration runs.**

Index Terms—analog front end, automatic calibration, DC offset, geophone, mixed-signal systems, programmable system-on-chip, PSoC

EXPERIMENTO/RESULTADO PENDIENTE: ESTE ES UN BORRADOR INTERNO. Antes de enviar se deben completar los valores cuantitativos, reemplazar las cajas de figuras y eliminar todas las anotaciones rojas y amarillas. El artículo definitivo ya está escrito en inglés, como exige URUCON.

I. INTRODUCTION

Multichannel surface-wave methods use spatially distributed vibration measurements to estimate the dispersive behavior of Rayleigh waves and infer near-surface mechanical properties [1], [2]. Although the processing methods are well established, the acquisition hardware remains a significant part of the cost and deployment complexity. Compact wireless nodes built from inexpensive geophones and mixed-signal microcontrollers are therefore attractive for dense and reconfigurable arrays [3].

The present work forms part of a low-cost Internet-of-Things acquisition platform based on SM-24 moving-coil geophones. A PSoC 5LP was selected as the analog front end because it integrates programmable-gain amplifiers, operational

amplifiers, analog filters, voltage DACs, analog multiplexers, a delta-sigma ADC, a digital filter block, DMA, and an Arm Cortex-M3 processor in a single device [4]. In the proposed implementation, the same ADC and digital filter are time-shared between geophone acquisition and front-end calibration. This resource reuse avoids a separate calibration converter or processor. The same reconfigurability that makes the device attractive also imposes a limitation: its internal analog blocks cannot be selected independently for low offset, low drift, and low flicker noise. Lopin and Lopin reported the same tradeoff in a single-chip PSoC potentiostat, noting substantially poorer offset and voltage-noise specifications than those of purpose-selected external amplifiers [5].

The geophone conditioning circuit used in this work is derived from the cascade correction and subtraction topology proposed by Ma *et al.* [6]. When mapped to the PSoC, accumulated DC errors prevented the use of sufficiently aggressive gain settings and became particularly severe at the ADDER output.

A continuously active DC servo was implemented and experimentally evaluated as an initial solution. Its performance exposed an unavoidable compromise: a very low servo bandwidth preserved the low-frequency vibration signal but required an impractically long centering time, whereas a faster loop disturbed the signal band. Moreover, every programmable-gain change altered the loop conditions and initiated a new transient. The proposed method therefore performs a finite foreground calibration before measurement and disconnects the controller afterward.

The contribution is threefold: 1) a per-stage analog calibration architecture that reuses the PSoC analog multiplexer, ADC, digital filter, and four VDACs; 2) a controller-independent formulation, demonstrated with a gain-aware PI implementation using deadband, anti-windup, slew limiting, stability detection, and one-LSB refinement; and 3) a measurement-gated operating policy that freezes all corrections before seismic data are recorded. Consequently, the calibration loop introduces no additional closed-loop poles into the acquisition transfer function.

II. PLATFORM DESIGN AND CALIBRATION MOTIVATION

A. Processing Architecture

The platform was selected according to five criteria: low relative cost, reconfigurable analog routing, hardware-assisted signal processing, reuse of acquisition resources across operating modes, and sufficient conversion resolution for both measurement and internal diagnostics. In acquisition mode, the PSoC conditions and digitizes the geophone output. In calibration mode, the input chain remains unchanged, but the observation path, digital-filter coefficients, and DAC-control firmware are reconfigured to close a temporary mixed-signal loop.

Self-calibration using on-chip resources is not itself new. The PSoC-Stat injects five known current levels with an IDAC and uses the resulting ADC values to estimate the gain and offset of a TIA/ADC signal chain [5]. Infineon's AN68403 applies an internal VDAC to a PGA/ADC chain, measures the system offset with grounded inputs, and stores gain and offset corrections in EEPROM [7]. These methods primarily correct the numerical transfer characteristic observed at the ADC output. The objective here is different: the proposed loop changes the physical reference of every amplification stage so that each intermediate analog node returns to its intended operating point.

The available ADDER made it possible to implement a fixed DC injection as an early workaround. This solution was viable and functional for this particular circuit, but it is not generally transferable to front ends without an explicit summing node. More importantly, it is only a patch: earlier stages continue to drift before the correction point, while subsequent stages introduce new offsets after it.

For the i th stage, its slowly varying output can be represented as

$$y_i^{\text{DC}} = A_i y_{i-1}^{\text{DC}} + b_i + K_i u_i, \quad (1)$$

where A_i is the stage gain, b_i contains the amplifier and reference errors, K_i is the signed transfer from the trim DAC to the observed node, and u_i is the DAC code. Adjusting only the ADDER reference can force one selected output toward its desired value, but it does not constrain the remaining y_i^{DC} values. If an earlier stage reaches a rail, the summing correction cannot reconstruct the lost waveform; offsets appearing after the ADDER are also left uncompensated.

The proposed architecture instead chooses u_i independently so that every observed y_i^{DC} approaches its stage reference. In the differential GEO implementation, all four calibration targets correspond to zero ADC counts.

III. SELF-CALIBRATING GEOPHONE PLATFORM

A. Signal Conditioning and Observation Circuit

The final GEO prototype contains four calibrated points, denoted PGA, BP, ADDER, and LP. Each point has an independent 8-bit VDAC. An analog multiplexer connects one point at a time to the same 18-bit delta-sigma ADC used to digitize the geophone signal. Its readings are rescaled into VDAC-code-equivalent units, allowing the acquisition

converter to serve as the error sensor without adding a second ADC. Software-enforced mutual exclusion prevents two analog outputs from being connected simultaneously.

The ADC observation path includes a switched capacitor behind the selected analog-multiplexer MOSFET. The switch on-resistance acts as the resistive element of a limited RC anti-alias and smoothing network. This placement also isolates the operational-amplifier output from a direct capacitive load that could compromise stability. The PSoC digital filter block is also reused. During normal operation it implements the geophone acquisition filter; during calibration, when that processing path would otherwise be idle, its coefficients are replaced by an aggressive low-pass response that provides a stable estimate of the DC level. The estimator could be implemented in firmware, but hardware reuse reduces CPU intervention and preserves the existing acquisition architecture. At the end of calibration, the acquisition coefficients and signal path are restored.

B. Digitally Assisted Control Calibration

For each stage, the filtered observation $\hat{y}_i[n]$ produces the ADC-domain error

$$e_i[n] = r_i - \hat{y}_i[n], \quad (2)$$

where $r_i = 0$ for the current differential GEO configuration. The error is first mapped to the 8-bit DAC domain:

$$\varepsilon_i[n] = \text{round}\left(\frac{e_i[n] V_{\text{ADC}} N_{\text{DAC}}}{N_{\text{ADC}} V_{\text{DAC}}}\right), \quad (3)$$

with $V_{\text{ADC}} = 5000$ mV, $N_{\text{ADC}} = 262144$, $V_{\text{DAC}} = 4080$ mV, and $N_{\text{DAC}} = 255$ in the implemented configuration.

The calibration architecture requires only that the selected stage be observable and that the trim DAC provide sufficient control range. The controller can therefore be selected according to the plant and application requirements. In this prototype, a positional PI law was used because it is computationally simple:

$$u_i^*[n] = u_{i,0} + s_i \left[\frac{k_{p,i}}{K_i} \varepsilon_i[n] + \frac{k_{i,i}}{K_i} \sum_{q=0}^n \varepsilon_i[q] \right], \quad (4)$$

where $u_{i,0}$ is the nominal feedforward code, s_i accounts for the injection polarity, and K_i is the signed physical gain from a VDAC variation to the selected ADC node. For the PGA stage, K_i is calculated from the currently programmed gain; the remaining stage gains are stored in the calibration table.

The command is saturated to the valid VDAC interval and limited to one code per processed sample. Conditional integration provides clamping anti-windup. The deadband is

$$\Delta_i = \max(1, \lceil 1.2 |K_i| \rceil) \quad (5)$$

in equivalent DAC codes. Convergence requires the error to remain within the same quantized bucket for a configurable number of samples. A final trial at the adjacent VDAC code is retained only if it reduces the absolute filtered error. More elaborate control laws could be substituted without changing the observation and correction hardware.

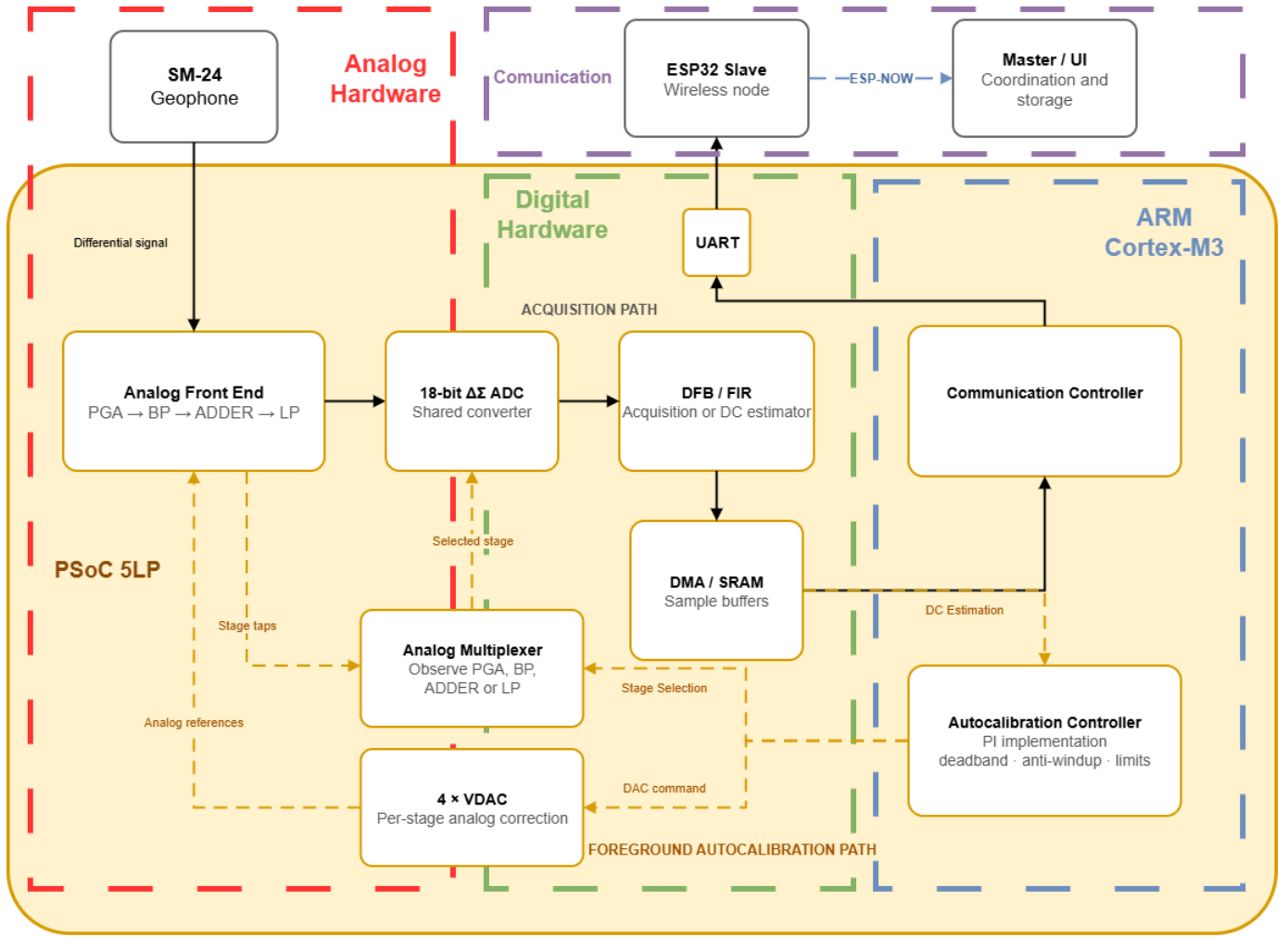


Fig. 1. Proposed analog front end and digitally assisted calibration path.

The 8-bit VDAC resolution is sufficient for the offset range and residual-error target of the present prototype. If a different application requires a smaller correction step, a PSoC IDAC and a resistor can generate a tailored voltage range; buffered and bidirectional topologies can also provide finer effective VDAC resolution [8]. This is an implementation option rather than a limitation of the calibration principle: the controller, correction range, and step size can be adapted to the expected drift while retaining the same observe–estimate–correct sequence.

C. Embedded Sequence and Persistence

At boot, after a gain change, or upon command, the firmware loads the stored DAC values for the selected PGA setting and verifies the resulting stage offsets. If every stage is already within tolerance, the measurement path is restored without executing a complete calibration. Otherwise, the controller processes PGA, BP, ADDER, and LP sequentially. Each stage is allowed to settle for a configurable number of samples after multiplexer switching and FIR-history reset. A watchdog and a per-stage sample limit prevent an indefinite calibration.

Successful codes can be stored in EEPROM. Nine records correspond to the nine programmable PGA settings. Each record contains the hardware class, gain code, stage count, valid-stage mask, four DAC codes, format version, and CRC. Under unchanged operating conditions, changing gain therefore usually requires only loading and verifying a previously calibrated code set. If verification fails because the operating conditions or drift have changed, the complete sequence runs automatically. A failed or incomplete run is not allowed to overwrite a previously valid record.

During acquisition the PI is inactive, the observation multiplexer is returned to the LP output, and all VDAC codes remain constant. Thus, the method changes the DC operating point but does not introduce a time-varying feedback loop into the acquired signal.

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PSoC Foreground Per-Stage Calibration

Controller-independent architecture; PI implementation shown

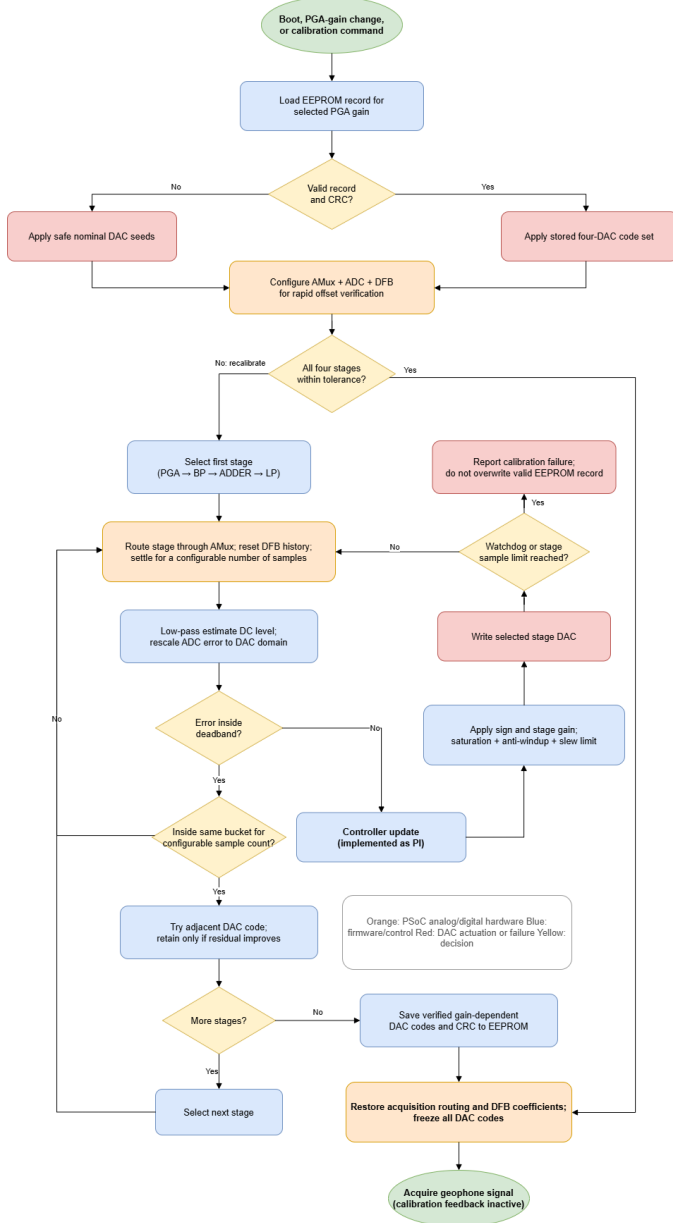


Fig. 2. Flowchart of the proposed foreground autocalibration algorithm, including stored-code verification, sequential per-stage correction, and transition to acquisition.

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